

AI Usage DISCLOSURE FORM

1. Team Details

Team Name: Srm_Developers

Project / Product Name: Smart Guided Troubleshooting Engine

Organization / Institution (if any): SRM Institute of Science and Technology (SRM IST)

Submission Date: 30 September 2026

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project? YES

Yes. AI was used as a development and engineering-assistance tool for problem understanding, code generation/refactoring, UI/UX work, documentation, testing, debugging, and integration. The team reviewed, modified, integrated, and tested the generated outputs.

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming: YES — AI assistance was used to explore and refine the troubleshooting-engine concept, pipeline architecture, feature ideas, and integration approach.

Code generation or assistance: YES — AI coding assistance was used for implementation, refactoring, adapter integration, API development, frontend development, and test development.

UI / UX design: YES — AI assistance was used to develop and refine the Samsung Galaxy One UI-inspired frontend structure, interaction flow, visual hierarchy, and styling.

Content creation: YES — AI assistance was used to draft technical documentation, README content, project descriptions, and explanatory material.

Data analysis: YES — AI assistance was used to inspect project data, retrieval outputs, test results, repository structure, and integration behavior.

Testing / debugging: YES — AI assistance was used to diagnose errors, inspect logs, improve tests, debug integration issues, and verify expected behavior.

Other: YES — AI assistance was used for repository auditing, architecture review, integration planning, code review, and development workflow support.

4. Feature Origin Classification (Self-Generated / AI-Generated / Both)

1. Feature Name: Natural-Language Troubleshooting / Query Understanding

2. Self-Generated/AI-Generated/Both: Both

3. Description: AI tools/platform used: Google Gemini integration and AI coding assistance. Prompt used: instructions to interpret vague mobile-device complaints and convert them into structured troubleshooting/problem-understanding information. Output Summary: normalized diagnostic intent and structured problem-understanding data. Modification: the team reviewed the output and integrated it with project schemas, query enrichment, retrieval, orchestration, and deterministic validation.

1. Feature Name: Retrieval-Augmented Knowledge Retrieval

2. Self-Generated/AI-Generated/Both: Both

3. Description: AI tools/platform used: AI coding assistance, SentenceTransformers and FAISS-based retrieval components. Prompt used: instructions to ground troubleshooting in approved knowledge rather than invent unsupported solutions. Output Summary: relevant knowledge-base records for the user complaint. Modification: the team integrated the retrieval module through an adapter, added fallback matching, and connected it to the orchestrator.

1. Feature Name: Grounded Troubleshooting Plan Generation

2. Self-Generated/AI-Generated/Both: Both

3. Description: AI tools/platform used: Google Gemini and AI coding assistance. Prompt used: instructions to generate a structured troubleshooting plan using retrieved context and avoid unsupported actions when grounded context is unavailable. Output Summary: structured troubleshooting actions and steps. Modification: the team constrained the planning stage with project schemas and deterministic post-processing.

1. Feature Name: Deeplink Validation and Action Ordering

2. Self-Generated/AI-Generated/Both: Both

3. Description: AI tools/platform used: AI coding assistance. Prompt used: instructions to integrate a deterministic deeplink catalog validator and action-ordering layer. Output Summary: validated catalog-backed deeplinks, normalized categories, and ordered actions. Modification: the team integrated the validator and mapper, rejected unverified links, and removed deeplinks from manual actions.

1. Feature Name: Fast-Path Cache

2. Self-Generated/AI-Generated/Both: Both

3. Description: AI tools/platform used: AI coding assistance. Prompt used: instructions to implement a lightweight query cache for repeated troubleshooting requests. Output Summary: SHA-256 keyed in-memory cache with TTL and hit/miss behavior. Modification: the team integrated the cache into the application pipeline and tested cache-hit behavior.

1. Feature Name: REST API and End-to-End Orchestration

2. Self-Generated/AI-Generated/Both: Both

3. Description: AI tools/platform used: AI coding assistance and architecture-review assistance. Prompt used: instructions to connect the AI, retrieval, validation, cache, and frontend modules through a FastAPI REST API. Output Summary: integrated /health and /troubleshoot endpoints with dependency-injected adapters and canonical response models. Modification: the team reviewed, corrected, tested, and integrated the generated implementation into the project.

1. Feature Name: Samsung One UI-Inspired Frontend

2. Self-Generated/AI-Generated/Both: Both

3. Description: AI tools/platform used: AI coding/design assistance. Prompt used: instructions to design a Samsung Galaxy One UI-inspired troubleshooting interface with diagnostic states, interactive steps, progress tracking, and clear action cards. Output Summary: React/TypeScript frontend components and styling. Modification: the team reviewed the design, adjusted the components and styling, connected the API, and tested the user flow.

1. Feature Name: Project Documentation and README

2. Self-Generated/AI-Generated/Both: Both

3. Description: AI tools/platform used: AI writing and repository-audit assistance. Prompt used: instructions to inspect the integrated project and prepare comprehensive technical documentation. Output Summary: architecture, setup, API, testing, deployment, limitations, and workflow documentation. Modification: the team reviewed the generated documentation and adapted it to the project's actual implementation.

5. Ethical & Compliance Confirmation

AI usage complies with guidelines and policies: YES

No proprietary or copyrighted data was intentionally misused: I AGREE

The team reviewed and modified AI-assisted outputs before incorporating them into the project.

6. Declaration & Sign-Off

Name of Team Representative: Bharath PVS

Role: Team Representative / Integration Lead

Signature:  _____

Date: 30 September 2026